The "switch" statement

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The "switch" statement

A switch statement can replace multiple if checks.
It gives a more descriptive way to compare a value when multiple variants.

The syntax:

The switch has one or more case blocks and an optional default. It looks like this:

```
switch(x){
   case 'value1': // if (x == 'value1')
      …
      [break]
   case 'value2': // if (x == 'value2')
      …
      [break]
   …
   default:
      …
```

- The value of X is checked for a strict equality to the value of the first case (that is, value1) then to the second (value2) and so on.
- If the equality is found, switch starts to execute the code starting from the corresponding case. Until the nearest break (or until the end of the switch)
- If no case is matched, then the default code is executed (if it exists)

```
const prompt = require('prompt-sync')();
let a = Number(prompt(`Enter a number: `));
switch(a){
    case 3:
        console.log(`Too small`);
        break;
    case 4:
        console.log(`Exactly!`);
        break;
    case 5:
        console.log(`Too big`);
        break;
    default:
        console.log(`I don't know such values`)
}
```

Output:

```
PS C:\Users\bluej\Documents\Student
s\DeeptaNarayanSarkar\College\Secon
dSem\WebDev\Git\Coding\Project0> no
de JS/Switch/switchone.js
Enter a number: 4
Exactly!
PS C:\Users\bluej\Documents\Student
s\DeeptaNarayanSarkar\College\Secon
dSem\WebDev\Git\Coding\Project0> no
de JS/Switch/switchone.js
Enter a number: 5
Too big
PS C:\Users\bluej\Documents\Student
s\DeeptaNarayanSarkar\College\Secon
dSem\WebDev\Git\Coding\Project0> no
de JS/Switch/switchone.js
Enter a number: 6
I don't know such values
PS C:\Users\bluej\Documents\Student
s\DeeptaNarayanSarkar\College\Secon
dSem\WebDev\Git\Coding\Project0> no
de JS/Switch/switchone.js
Enter a number: 3
Too small
PS C:\Users\bluej\Documents\Student
s\DeeptaNarayanSarkar\College\Secon
dSem\WebDev\Git\Coding\Project0>
```

Here the switch starts to compare a from the first case variant, that is 3. The match fails. Then 4. That's a match, so the execution starts from case four until the nearest break. If there is no break, then the execution continues with the next case without any checks. An example without break:

```
const prompt = require('prompt-sync')();
let a = Number(prompt(`Enter a number: `));
switch(a){
    case 3:
        console.log(`Too small`);

    case 4:
        console.log(`Exactly!`);

    case 5:
        console.log(`Too big`);

    default:
        console.log(`I don't know such values`)
}
```

Output:

```
Enter a number: 4
Exactly!
Too big
I don't know such values
```

Any expression can be switch/case argument
Both switch and case allow arbitrary expression, for example:

```
let a = "1";
let b = 0;
switch (+a) {
    case b + 1:
        console.log("this runs, because +a is 1, exactly equals b+1")
        break;
    default:
        console.log(`this doesn't run`)
        break;
}
```

Output:

```
this runs, because +a is 1, exactly equals b+1
```

Grouping of "case"
Several variants of case which share the same code can be grouped.
For example, if we want the same code to run for case 3 and case 5.

```
let a = 3;
switch (a) {
    case 4:
        console.log(`Right`);
        break;
    case 3: // (*)grouped two cases
    case 5:
        console.log(`Wrong!`)
        console.log(`why don't you take a math class?`)
        break
    default:
        console.log(`The result is strange. Really!`)
        break;
}
```

Output:

```
   Wrong!
   why don't you take a math class?
```

Now both 3 and 5 show the same message. The ability to group cases is a side effect of house, which / case works without brake. Here the execution of case three starts from the line (*). And goes through case 5, because there is no break.

Type matters

Let's emphasize that equality check is always strict. The values must be of the same type to match. For example, let's consider the code:

```
const prompt = require('prompt-sync')();
let a = prompt(`Enter a value: `);
switch(a){
    case `0`:
    case `1`:
        console.log(`One or zero`)
        break;
    case `2`:
        console.log(`Two`)
        break
    case 3:
        console.log(`Never execute`);
        break;

    default:
        console.log(`I don't know such values`)
}
```

Output:

```
● de JS/Switch/switchfour.js
  Enter a value: 0
  One or zero
  PS C:\Users\bluej\Documents\Student
  s\DeeptaNarayanSarkar\College\Secon
  dSem\WebDev\Git\Coding\Project0> no
● de JS/Switch/switchfour.js
  Enter a value: 2
  Two
  PS C:\Users\bluej\Documents\Student
  s\DeeptaNarayanSarkar\College\Secon
  dSem\WebDev\Git\Coding\Project0> no
● de JS/Switch/switchfour.js
  Enter a value: 3
  I don't know such values
```

1. For 0, 1, one, The first console.log() runs.

2. For 2 the second console.log() runs.
3. But for 3, the result of the prompt is a string "3". Which is not strictly equal === to the number 3. So we have got a dead code in case three. The default variant will execute.